

TCR Additions Playbook

Addition-specific practices that layer on top of the TCR General Knowledge document. Covers what an addition needs that other job types do not — permitting realities, long-lead structural items, retrofit patterns that almost always appear, and the change-order risks unique to tying new construction into existing structure.

1. When to Use This Document

Apply this playbook when the project scope describes **adding new square footage to an existing home** — bedroom addition, bathroom addition, master-suite addition, second-story addition, or a garage-to-living conversion.

For pure interior remodels of existing space, use the General Knowledge document baseline without this layer.

2. Septic — the Silent Long-Lead Item

For additions that add a bedroom, a bathroom, or any other fixture that the TDEC (Tennessee Department of Environment and Conservation) counts against septic capacity, the septic permit process is the **longest single lead time in the project** and often determines the realistic on-site start date.

Septic work is only needed if it is specifically called out in the scope.

The TDEC process (~6 weeks total)

1. Initial contact with TDEC.
2. Soil scientist site test — \$500 permit fee plus scientist's time.
3. Soil report completed by TDEC.
4. TDEC review and permit issuance.

End-to-end this is typically **4–8 calendar weeks** from first contact to permit-in-hand. Treat it as a `lead_time` task in `pre_construction` with **lead_time_days: 70 minimum**.

3. Structural Long-Lead Items

Two structural items consistently drive the addition critical path. Both belong in `pre_construction` as procurement tasks gating the relevant framing tasks via FS.

Roof trusses (custom)

4–6 weeks lead time from order to delivery. Order at structural sign-off, well before framing starts. Trusses can land weeks before they are installed without issue. Use **lead_time_days: 35** as the default.

LVL beams

Nominal lead time is 7–14 days, but **default to 21 days** for safety when an LVL is on the critical path. LVL availability fluctuates with lumber market conditions and "next week" can quietly become "three weeks" without notice. Better to early-order and have the beam waiting than to halt framing.

4. Windows — Order at Permit Submittal

Windows are ordered a **minimum of 21 calendar days** prior to demo (the first phase of the project), not at framing start. Custom windows (28–42 days) usually arrive just-in-time.

Verify with the customer whether windows are a **custom** or **standard** order item for the project — the lead time doubles the difference.

5. Retrofit Detection — Almost Always Present

It is rare for an addition project to be only new construction. The addition almost always disturbs existing space:

- The wall where the new addition ties in is usually load-bearing.
- New bedrooms often trigger existing-bath modifications for code or layout reasons.
- New master suites often involve closing off or repurposing an existing hallway or closet.

How to handle a retrofit section

When a breakdown section describes work on an area that is NOT the addition objective, treat it as retrofit:

1. Put it in its **own Component**, not commingled with the new-construction Components.
2. Do **not** gate retrofit work on new-construction dependencies that do not physically apply (a hall bath modification is not waiting on new exterior framing of a bedroom addition).
3. Retrofit work runs in parallel with new construction wherever trade and crew constraints allow.
4. Emit a warning naming the retrofit section and what tipped you off — for example: "section X mentions hallway bath; addition objective is bedroom — flagging as retrofit."

Also, suggest a retrofit-items list to the PM. After the PM confirms, ask if there is any other work not specifically listed before finalizing the retrofit items, and where each should be completed during the project.

Common patterns to watch for

- **New bedroom addition** — existing hall bath modified for layout or code.
- **New master suite** — existing master closet repurposed or hallway reframed.
- **Second-story addition** — existing first-floor electrical panel upgraded for new load.

- **Garage-to-living conversion** — existing HVAC or panel modified to feed the new conditioned space.

6. Customer-Requested Early Items

It is common for the client to request a small piece of work be done **before** the main addition starts (a shower swap, a leaky-fixture fix, a quick demo) so they can live with it during the months of construction. These appear as small sections in the breakdown.

Verify whether each is critical-path or not. Treat them as standalone tasks scheduled in pre_construction or very early in the on-site timeline. Verify whether each gates on permitting, framing, or any addition-specific predecessor. Surface in the assumptions field when unsure whether a small section is a customer-requested early item versus part of the addition scope.

7. Concealed Roof Tie-In — Change-Order Risk

When the addition's new roof ties into the existing roof, the structural conditions above the existing ceiling are **concealed until demo opens them up**. Existing rafters or trusses may not be sized to accept the new load, may have hidden damage, or may be framed in a way that does not permit a clean tie-in.

Whenever a section describes roof tie-in to existing structure, emit a warning naming concealed-condition risk and reserve a **2–3 day schedule buffer** immediately after the concealed area is opened up. The buffer absorbs the discovery-and-redesign cycle that may follow.

8. Existing Electrical Service — Amperage Check Before Subpanel

If the addition's electrical scope includes a subpanel feeding the new space, the existing main service amperage **must be verified**. Most homes have a 200A service, but often do not have space for the subpanel without a service upgrade.

The amperage check is a **0.5-day task** in pre_construction (general trade). Its result drives whether the project needs an added service-upgrade scope.

If the breakdown includes a subpanel but no service-upgrade section, emit a warning: "subpanel scope present, no existing-service amperage verification task — confirm main service capacity before signed contract."

9. LVL Temporary Shoring — Independent Retrofit

When the project requires an LVL beam to replace an existing load-bearing wall (very common in additions that open the existing floor plan to the new space), the temporary shoring + LVL install + permanent shoring removal is **its own sub-chain**. It is gated by demo (the wall has to be exposed) but is NOT gated by new-construction framing.

Sequence

1. demo.expose_bearing_wall — 1 day, demo trade.
2. structural.temporary_shoring — 0.5 day, framing trade.
3. structural.install_lvl — 1–2 days, framing, FS after procurement.lvl.
4. structural.remove_shoring — 0.5 day, framing.
5. inspect.framing covers this work along with the rest of the new framing.

This sub-chain can run in parallel with new exterior framing. Do not gate new framing on it unless the new floor or roof load actually bears on the LVL location.

10. Material Ordering — Interior Finishes Wait

The general "order tied to use date" rule is especially important for additions because there is a substantial gap (often 4–6 weeks) between project start and the interior finish phase, taken up by foundation, framing, and roof dry-in.

Interior finish materials (tile, cabinets, fixtures, flooring) are ordered **after the structural shell is underway**, not at project start.

Exceptions to the "wait" rule

- **Long-lead cabinets** (42–56 days) may need to be ordered at structural sign-off to hit the cabinet install date.
- **Long-lead tile** (custom, 14–21 days) ordered about 3 weeks before tile install starts.

Everything else (paint, flooring, trim, standard fixtures) is to be on site 2 weeks ahead of use date. Typical delivery dates for these standard items are approximately 1 week. This means we need to order roughly 3 weeks prior to starting that work.

11. Addition-Specific Phase Additions

Insert these on top of the General Knowledge phase spine where the scope calls for them:

- **pre_construction.tdec_septic** — when a septic-impacting fixture is added.
- **pre_construction.amperage_check** — when a subpanel is in scope.
- **structural_and_shell.roof_tie_in** — its own Component, with the concealed-condition buffer attached.

12. Mandatory Addition Warnings

In addition to the base warnings from the General Knowledge document, surface these when the conditions apply:

1. Roof tie-in to existing structure is in scope, but no schedule buffer is reserved after the concealed area is opened up.
2. Subpanel is in scope, but no existing-service amperage verification task is present.
3. New-construction sections and probable retrofit sections are mixed into the same Component.
4. Truss or LVL procurement is missing from pre-construction even though the breakdown includes framing that requires them.

13. Will's Walkthrough — the Closer

For every home-addition job, add a final checkpoint at the very end of the schedule: **Will's Walkthrough**. One working day after the previous final closeout task, Will visits the home, meets with the customer, and closes the job with a personal final walkthrough.

Model this as the last task in closeout, running the working day after the previous final closeout task.

14. Related Reference Materials

Companion artifacts within PM_audit that inform this playbook:

- TCR General Knowledge — the cross-job baseline this document layers on top of.
- 308_Scope.txt and 308_Evergreen_Addition_Breakdown.csv — an in-progress reference addition project.
- 308_Wills_Full_Audit_Transcript.txt — the source recording from Will used to validate and expand these rules.

Additional job-type documents (Bathrooms, Kitchens, Whole-Home Remodels) will follow the same structure — general behavior lives in the General Knowledge doc; the job-type layer captures only what is specific.